# Vanagarouthu Sree Chaitran

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## Education

**National Institute of Technology, Delhi** **Aug. 2025 – May. 2027**
*M.Tech in Mathematics and Computing* *GPA: 7.25/10*

**Vellore Institute of Technology, Vellore** **Sep. 2020 – May. 2024**
*B.Tech in CSE with IoT* *GPA: 7.86/10*

## Experience

**Artificial Intelligence Internship** **May 2023 – July 2023**
*Smart Bridge*

- Movie Recommendation Based on Emotion Using Web Scraping with IBM.
- Collected and processed large volumes of movie data from multiple online sources using Python-based web scraping techniques to support sentiment and emotion analysis.
- Developed an emotion-driven movie recommendation model integrating NLP and sentiment analysis, enabling personalized movie suggestions aligned with the user's emotional state.
- Designed an interactive Flask web interface to deliver real-time recommendations, improving user engagement and enhancing the overall recommendation experience.

## Projects

**Seasonal Arctic Sea-Ice Forecasting Using a Swin-UNet Deep Learning Model** | *GitHub*

- Built a complete Swin-UNet forecasting pipeline for Arctic sea-ice concentration, including an 18-channel spatiotemporal input (concentration, anomaly, climatology, thickness) on a 96×96 polar grid.
- Designed and trained a 60-model ensemble (20-fold × 3-seed) under a leakage-free temporal validation protocol to isolate the contribution of sea-ice-thickness inputs, using bootstrap resampling and FDR-corrected significance testing (p=0.031).
- Benchmarked forecast skill against CNN-based U-Net baselines at matched (15M) and larger (140M) parameter scales, isolating the effect of model capacity from architecture on prediction accuracy.

**Real-time Rendering for Cricket Shot Detection using Deep Learning** | *GitHub*

- Developed an automated system that can accurately detect and classify umpire gestures and cricket batting shots using deep learning techniques, specifically convolutional neural networks (CNNs) and long short-term memory (LSTM) networks, integrated with the MediaPipe framework for real time pose estimation.
- Achieved high accuracy rates of 84.34 Percent for classifying batting shots and 81.7 Percent for recognizing umpire gestures, demonstrating the potential for practical applications in cricket analysis, broadcasting, and coaching.

**Vector Database and Local RAG System Using C++ and Ollama** |

- Implemented a C++ vector database supporting HNSW, KD-Tree, and Brute Force similarity search with cosine, Euclidean, and Manhattan distance metrics.
- Integrated Ollama-based embeddings and local LLM generation to build a document-based RAG pipeline for semantic search and question answering.
- Exposed REST APIs for vector CRUD operations, benchmarking, document insertion, retrieval, and AI-powered responses, with a web UI for visualization and interaction.

## Journal Paper

Lane Prediction Optimization and Secure Traffic Management Using VANET **Harvest Journal, 2023**

## Technical Skills

**Languages**: C++, Python, MySQL, MATLAB, JavaScript, PHP, HTML, CSS
**Core CS**: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks
**AI/ML**: Machine Learning, Deep Learning, NLP, Vector Embeddings, RAG, Semantic Search
**Tools & Frameworks**: Git, VS Code, Google Cloud Platform, Flask, React, Ollama, REST APIs

## Achievements / Extracurricular

- Placement Coordinator, NIT Delhi **Jan 2026- May 2027**
- Core Committee Member – Nature Lovers Club, VIT Vellore **Jan 2023- Jan 2024**
- Achieved a rank of 6160 out of a competitive pool of 1,43,265 candidates in the TS EAMCET. **May 2020**